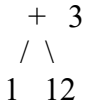
1. Starting from the following grammar for arithmetic expressions

   $Z \rightarrow E$,
   $E \rightarrow TC$,
   $C \rightarrow \varepsilon \mid +TC$,
   $T \rightarrow n \mid (E)$

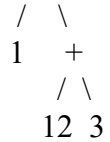
write a recursive-descent compiler for arithmetic expressions, which builds the expression tree (i.e. the abstract syntax tree, AST) for any arithmetic expression, then generates the code from the AST.
Example:

1+12+3




```
push 1
push 12
add
push 3
add
```

1+(12+3)




```
push 1
push 12
push 3
add
add
```